



Seashells

Royale

Hotel

Project done by:

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SEASHELLS HOTEL



Luxury Beach Holidays

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Your Tourist Agents

Seashells| Luxury Beach Holidays



Belinda Bentley
Tour Manager



Jordan Casnick
Senior Agent



Nadiya Rossman
Assistant Agent

SEASHELLS HOTEL DISCRIBTION:

Seashells Hotel is a 5 star Hotel with a nice view of the Mediterranean Sea. Being built in 5 floors and 20 rooms, we provide single and double rooms and you can also choose if you would like a room overlooking the beautiful blue Sea.





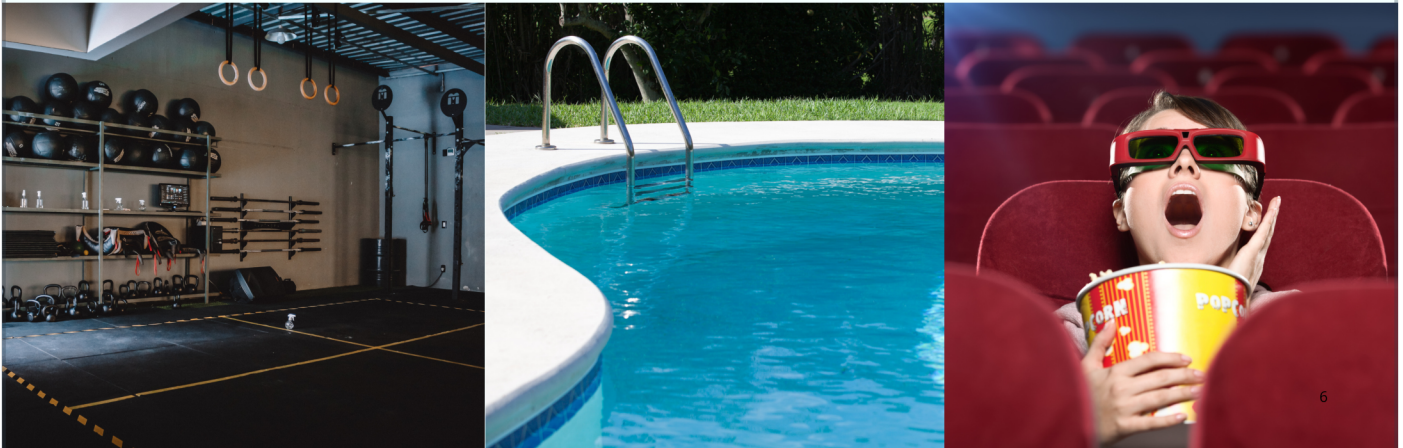
HOTEL SERVICES

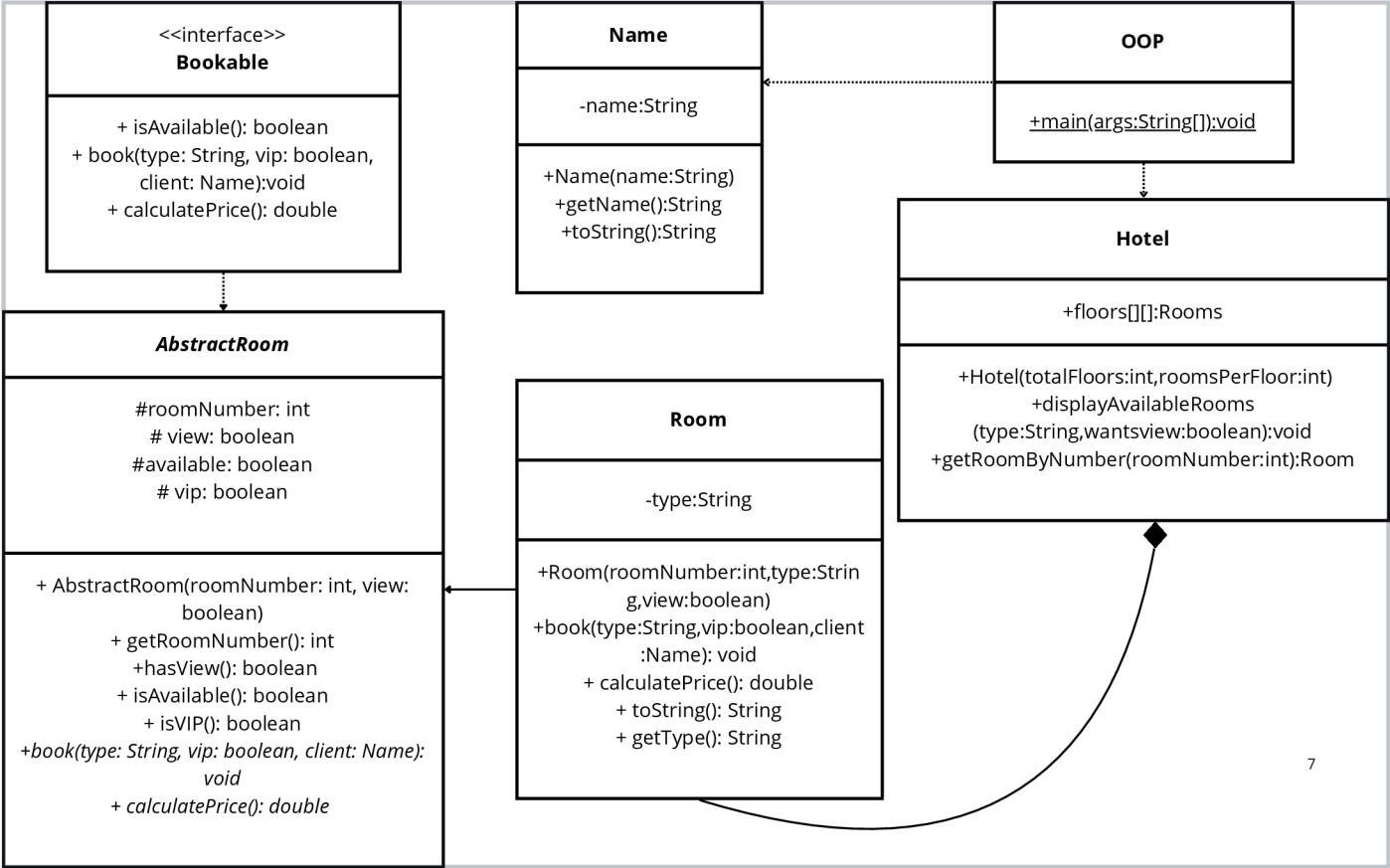
And to enjoy your Hotel Accommodation the Hotel services include: Free breakfast, hot line for room services and a Michelin star restaurant.

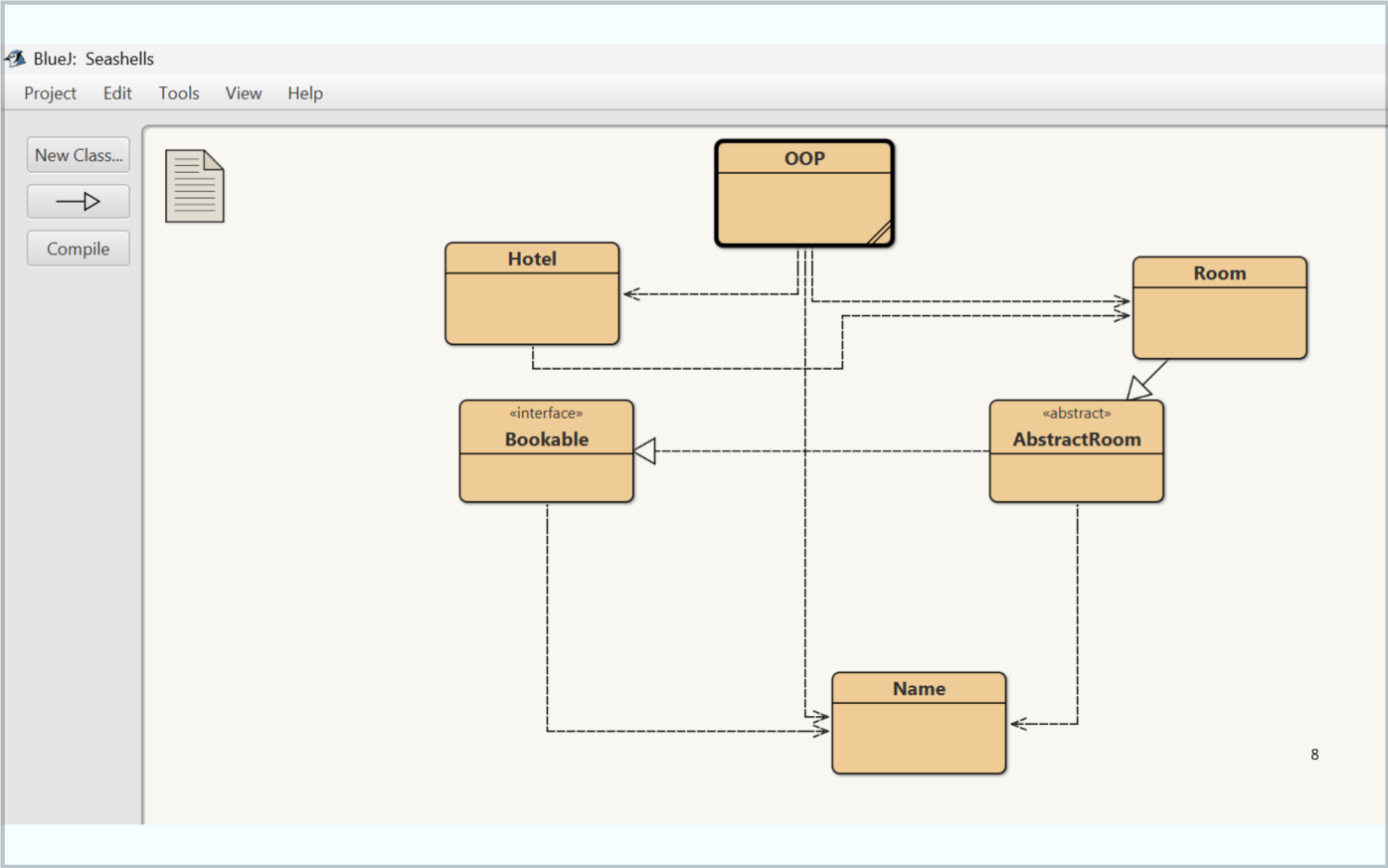
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THE VIP PACKAGE

And if you would like to book the VIP package you would have free access to : the gym, the pool , the spa and cinema.







```
import java.util.*;
public class OOP
{
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);

        Hotel hotel = new Hotel(5, 4);

        Room r501 = hotel.getRoomByNumber(501);
        r501.book("Single", true, new Name("Norah Ibrahim"));

        Room r402 = hotel.getRoomByNumber(402);
        r402.book("Double", false, new Name("Ali Mohammed"));

        Room r203 = hotel.getRoomByNumber(203);
        r203.book("Single", true, new Name("Somaia Adel"));

        Room r104 = hotel.getRoomByNumber(104);
        r104.book("Double", false, new Name("Omar Tareq"));

        System.out.print("Enter client name: ");
        Name client = new Name(scan.nextLine());

        System.out.print("Enter room type: ");
        String type = scan.nextLine();

        System.out.print("View or without view (yes/no): ");
        boolean wantsView = scan.nextLine().equalsIgnoreCase("yes");
    }
}
```

Class compiled - no syntax errors

saved

```
System.out.print("Enter room type: ");
String type = scan.nextLine();

System.out.print("View or without view (yes/no): ");
boolean wantsView = scan.nextLine().equalsIgnoreCase("yes");

System.out.print("Do you want VIP package? (yes/no): ");
boolean vip = scan.nextLine().equalsIgnoreCase("yes");

hotel.displayAvailableRooms(type, wantsView);

System.out.print("Enter the room number you want to book: ");
int selectedRoomNumber = scan.nextInt();

Room selectedRoom = hotel.getRoomByNumber(selectedRoomNumber);
if (selectedRoom != null && selectedRoom.isAvailable()
    && selectedRoom.getType().equalsIgnoreCase(type)
    && selectedRoom.hasView() == wantsView) {
    selectedRoom.book(type, vip, client);
    System.out.println("Total price: $" + selectedRoom.calculatePrice());
} else {
    System.out.println("Invalid room selection or room is not available.");
}
```

Class compiled - no syntax errors

```
public class Hotel
{
    public Room[][] floors;

    public Hotel(int totalFloors, int roomsPerFloor) {
        floors = new Room[totalFloors][roomsPerFloor];

        for (int i = 0; i < totalFloors; i++) {
            int floorNumber = i + 1;

            for (int j = 0; j < roomsPerFloor; j++) {
                int roomNumber = floorNumber * 100 + (j + 1);
                String type = (j == 0 || j == 2) ? "Single" : "Double"; // if j equals 1 or 3
                boolean view = (j == 0 || j == 1); //if j equals 1 or 2

                floors[i][j] = new Room(roomNumber, type, view);
            }
        }
    }

    public void displayAvailableRooms(String type, boolean wantsView) {
        System.out.println("\nAvailable Rooms for type: " + type + ", view: " + wantsView);
        boolean found = false;

        for (int i = 0; i < floors.length; i++) {
            for (int j = 0; j < floors[i].length; j++) {
                Room room = floors[i][j];
            }
        }
    }
}
```

Class compiled - no syntax errors

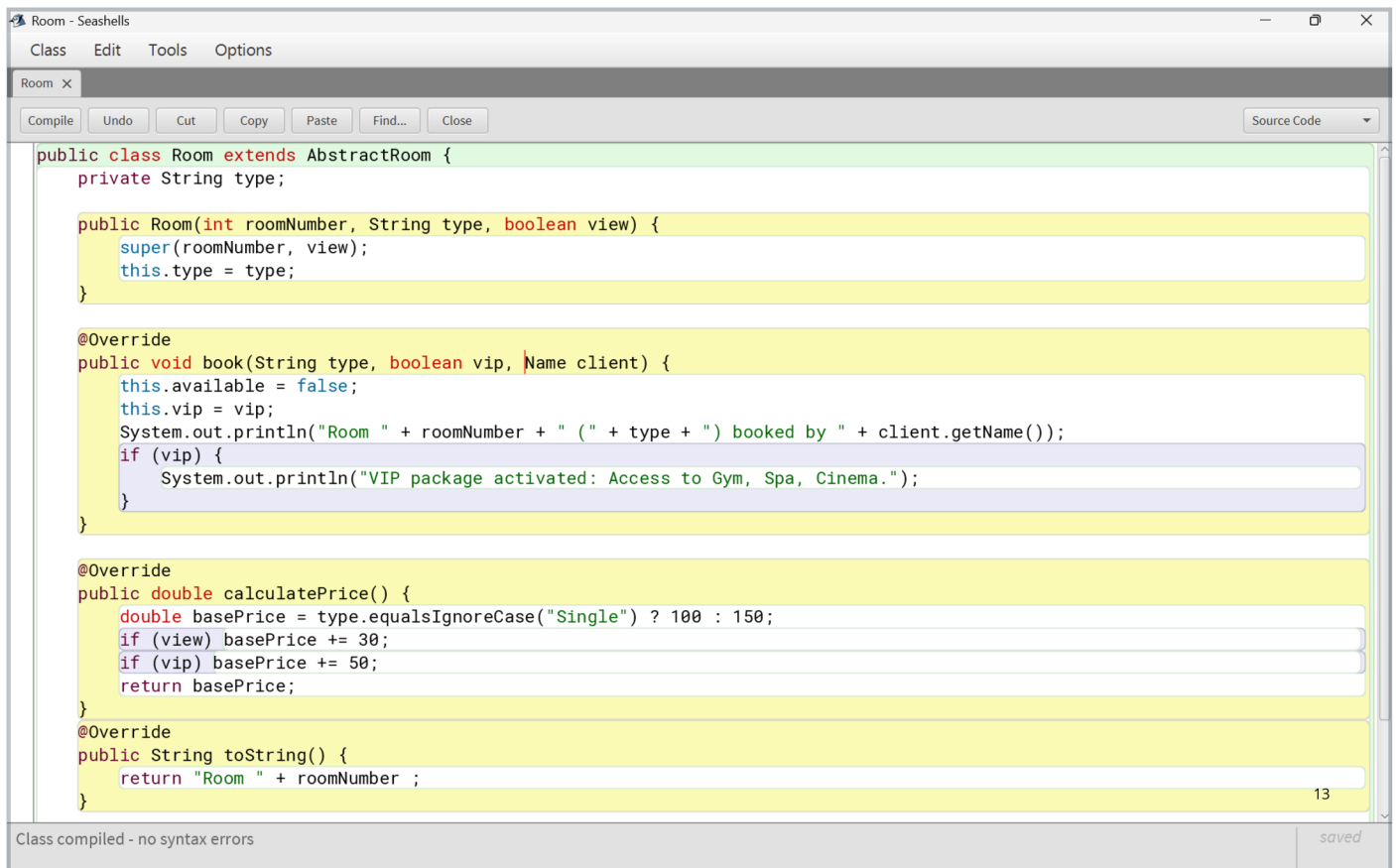
saved

```
Hotel - Seashells
Class Edit Tools Options
Hotel x
Compile Undo Cut Copy Paste Find... Close Source Code
for (int i = 0; i < floors.length; i++) {
    for (int j = 0; j < floors[i].length; j++) {
        Room room = floors[i][j];
        if (room.getType().equalsIgnoreCase(type)
            && room.hasView() == wantsView
            && room.isAvailable()) {
            System.out.println(room);
            found = true;
        }
    }
}

if (found==false) {
    System.out.println("No available rooms match your criteria.");
}

public Room getRoomByNumber(int roomNumber) {
    for (int i = 0; i < floors.length; i++) {
        for (int j = 0; j < floors[i].length; j++) {
            Room room = floors[i][j];

            if (room.getRoomNumber() == roomNumber) {
                return room;
            }
        }
    }
    return null;
}
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Class compiled - no syntax errors saved
```

```
public class Room extends AbstractRoom {
    private String type;

    public Room(int roomNumber, String type, boolean view) {
        super(roomNumber, view);
        this.type = type;
    }

    @Override
    public void book(String type, boolean vip, Name client) {
        this.available = false;
        this.vip = vip;
        System.out.println("Room " + roomNumber + " (" + type + ") booked by " + client.getName());
        if (vip) {
            System.out.println("VIP package activated: Access to Gym, Spa, Cinema.");
        }
    }

    @Override
    public double calculatePrice() {
        double basePrice = type.equalsIgnoreCase("Single") ? 100 : 150;
        if (view) basePrice += 30;
        if (vip) basePrice += 50;
        return basePrice;
    }

    @Override
    public String toString() {
        return "Room " + roomNumber ;
    }
}
```

Class compiled - no syntax errors

saved

Room - Seashells

Class Edit Tools Options

Room X

Compile Undo Cut Copy Paste Find... Close

Source Code

```
super(roomNumber, view);
this.type = type;
}

@Override
public void book(String type, boolean vip, Name client) {
    this.available = false;
    this.vip = vip;
    System.out.println("Room " + roomNumber + " (" + type + ") booked by " + client.getName());
    if (vip) {
        System.out.println("VIP package activated: Access to Gym, Spa, Cinema.");
    }
}

@Override
public double calculatePrice() {
    double basePrice = type.equalsIgnoreCase("Single") ? 100 : 150;
    if (view) basePrice += 30;
    if (vip) basePrice += 50;
    return basePrice;
}

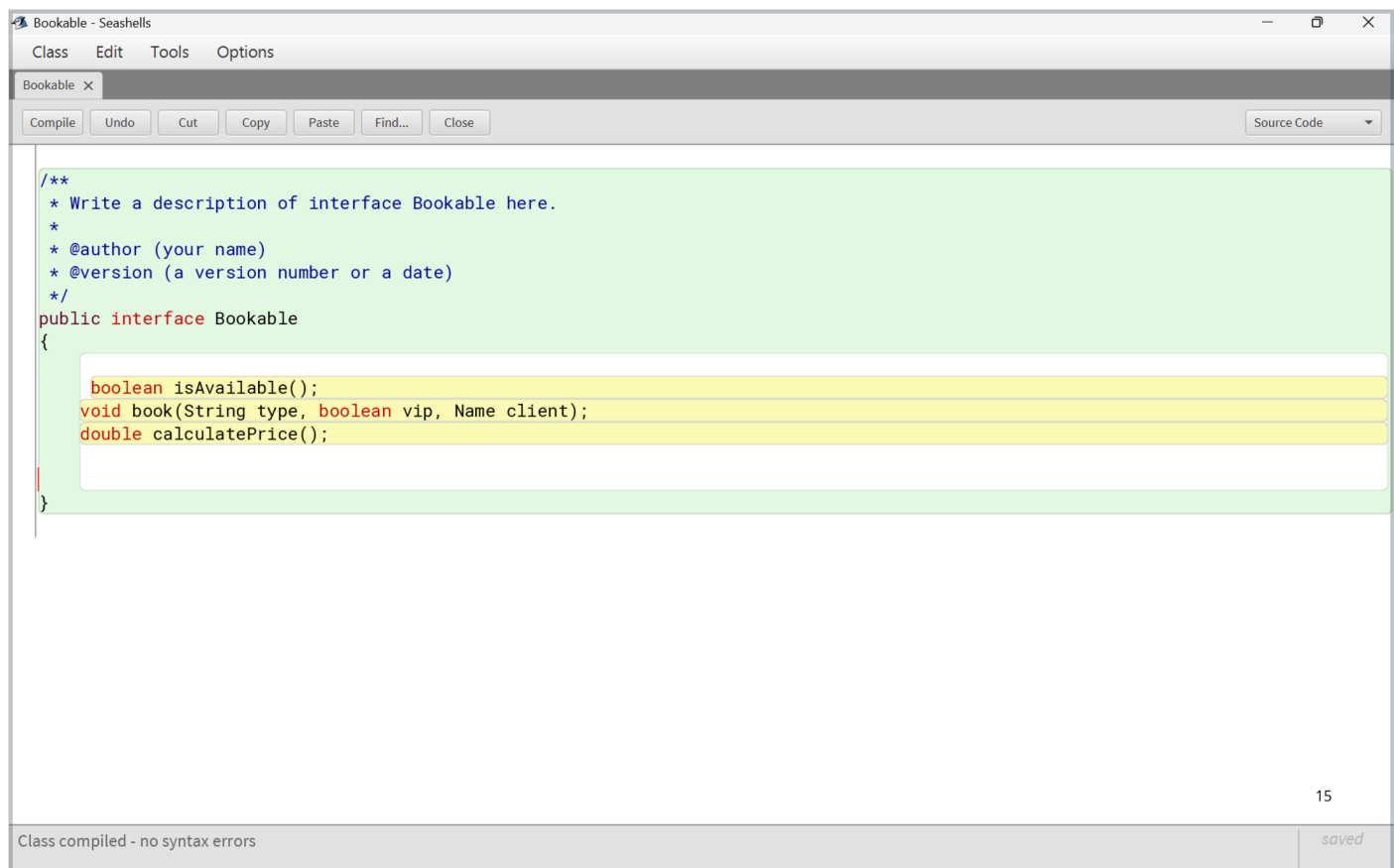
@Override
public String toString() {
    return "Room " + roomNumber ;
}

public String getType() {
    return type;
}
}
```

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Class compiled - no syntax errors

saved



AbstractRoom - Seashells

Class Edit Tools Options

AbstractRoom x

Compile Undo Cut Copy Paste Find... Close

Source Code

```
public abstract class AbstractRoom implements Bookable{
    protected int roomNumber;
    protected boolean view;
    protected boolean available = true;
    protected boolean vip = false;

    public AbstractRoom(int roomNumber, boolean view) {
        this.roomNumber = roomNumber;
        this.view = view;
    }

    public int getRoomNumber() {
        return roomNumber;
    }

    public boolean hasView() {
        return view;
    }

    public void setVIP(boolean vip) {
        this.vip = vip;
    }

    public boolean isVIP() {
        return vip;
    }

    public boolean isAvailable() {
        return available;
    }
}
```

Class compiled - no syntax errors

16 saved

AbstractRoom - Seashells

Class Edit Tools Options

AbstractRoom X

Compile Undo Cut Copy Paste Find... Close

Source Code

```
public int getRoomNumber() {  
    return roomNumber;  
}  
  
public boolean hasView() {  
    return view;  
}  
  
public void setVIP(boolean vip) {  
    this.vip = vip;  
}  
  
public boolean isVIP() {  
    return vip;  
}  
  
public boolean isAvailable() {  
    return available;  
}  
  
public void setAvailable(boolean available) {  
    this.available = available;  
}  
  
public abstract void book(String type, boolean vip, Name client);  
  
public abstract double calculatePrice();
```

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Class compiled - no syntax errors

saved

Name - Seashells

Class Edit Tools Options

Name X

CompileUndoCutCopyPasteFind...CloseSource Code

```
/**
 * Write a description of class Name here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Name
{
    private String name;

    public Name(String name) {
        this.name = name;
    }

    public void setName(String name){
        this.name=name;
    }

    public String getName() {
        return name;
    }

    @Override
    public String toString() {
        return name;
    }
}
```

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Class compiled - no syntax errors

saved

```
BlueJ: Terminal Window - Seashells
Options
Room 501 (Single) booked by Norah Ibrahim
VIP package activated: Access to Gym, Spa, Cinema.
Room 402 (Double) booked by Ali Mohammed
Room 203 (Single) booked by Somaia Adel
VIP package activated: Access to Gym, Spa, Cinema.
Room 104 (Double) booked by Omar Tareq
Enter client name: Sajeda
Enter room type: double
View or without view (yes/no): yes
Do you want VIP package? (yes/no): yes

Available Rooms for type: double, view: true
Room 102
Room 202
Room 302
Room 502
Enter the room number you want to book: 202
Room 202 (double) booked by Sajeda
VIP package activated: Access to Gym, Spa, Cinema.
Total price: $230.0

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Can only enter input while your program is running
```

Room 501 (Single) booked by Norah Ibrahim
VIP package activated: Access to Gym, Spa, Cinema.
Room 402 (Double) booked by Ali Mohammed
Room 203 (Single) booked by Somaia Adel
VIP package activated: Access to Gym, Spa, Cinema.
Room 104 (Double) booked by Omar Tareq
Enter client name: **Heba**
Enter room type: **single**
View or without view (yes/no): **view**
Do you want VIP package? (yes/no): **no**

Available Rooms for type: single, view: false
Room 103
Room 303
Room 403
Room 503
Enter the room number you want to book: **402**
Invalid room selection or room is not available.